

SUMMARY TABLE: ESSENTIAL MATHEMATICS GRADE 10

Sub-Strand 2.3: Trigonometry — Teacher Reference (Pre-filled)

SUMMARY TABLE: ESSENTIAL MATHEMATICS GRADE 10	
Sub-Strand	Sub-Strand 2.3: Trigonometry
Driving Question	DRIVING QUESTION: How can we use the angles and one side of a right-angled triangle to find any unknown height or distance in our environment — without ever measuring it directly?

INSTRUCTIONS

FOR TEACHERS: This is the teacher reference version — each row is pre-filled with expected student responses. Use it to assess student Summary Tables, identify gaps, and prepare discussion questions. The student version has blank cells for students to complete after each lesson.

Lesson # and Title	What did I observe?	What did I learn?	How does this explain the phenomenon?
Lesson 1 The Water Tank Problem: Predict and Name the Sides	We observed that the water tank tower at Kilimani Secondary School cannot be climbed, yet Wanjiru and Otieno stand at different distances from its base and sight the top of the tank to record different angles. The tower, the ground and the line of sight form a right-angled triangle.	We learned to name the sides of a right-angled triangle relative to a chosen acute angle θ : the hypotenuse is opposite the right angle and is the longest side; the opposite side is across from θ ; the adjacent side is next to θ but is not the hypotenuse. When θ moves to the other acute angle, opposite and adjacent swap but the hypotenuse stays the same.	We can say that the height of the tank above eye level is the opposite side and the distance from the learner to the tower is the adjacent side for the angle at the learner's eye, but we cannot yet explain how the angle and distance give the height, or whether Wanjiru and Otieno should get the same answer. These questions are on our Driving Question Board and we will answer them in the next lessons.
Lesson 2 Deriving sine, cosine and tangent: Why does the tank method work at any distance?	What did you measure and calculate for the three 35° triangles A, B and C? Record the adjacent, opposite and hypotenuse lengths, and the three ratios (opposite $\div$ hypotenuse, adjacent $\div$ hypotenuse, opposite $\div$ adjacent) for each triangle. What did you notice about the ratios as the triangles got bigger?	What are $\sin \theta$, $\cos \theta$ and $\tan \theta$ in terms of opposite, adjacent and hypotenuse? What are the calculator values of $\sin 35^\circ$, $\cos 35^\circ$ and $\tan 35^\circ$ in DEG mode, and how close are they to your measured ratios?	Why do the ratios stay the same for triangles of different sizes with the same angle? How does this idea help explain how Wanjiru (8 m, 37°) and Otieno (10 m, 31°) at Kilimani Secondary School can both find the same tank height without climbing the tower? What do you still need to learn to calculate that height?
Lesson 3 Complementary Angles: Why Does $\sin 20^\circ$ Equal $\cos 70^\circ$?	Complete the Lesson 3 table. For the pairs $20^\circ/70^\circ$, $30^\circ/60^\circ$, $35^\circ/55^\circ$ and $45^\circ/45^\circ$, we found that $\sin$ of one angle equals $\cos$ of the other (for example $\sin 20^\circ = \cos 70^\circ = 0.3420$ and $\sin 35^\circ = \cos 55^\circ = 0.5736$). At 45° the sine and cosine are equal (0.7071).	Two acute angles in a right-angled triangle add up to 90° and are called complementary. The side opposite one angle is the side adjacent to the other. So $\sin \theta = \cos(90^\circ - \theta)$ and $\cos \theta = \sin(90^\circ - \theta)$. Wanjiru's 37° at the tank goes with 53° at	Write in your own words why $\sin 35^\circ = \cos 55^\circ$, using the words opposite, adjacent and hypotenuse, and say how this helps us get closer to finding the height of the Kilimani Secondary School water tank without climbing it.

		the top of her line of sight.	
Lesson 4 Choosing the Right Ratio: Solving for Unknown Sides and the First Full Tank-Height Calculation	We observed that to find an unknown side, we label the triangle from the angle, pick the ratio that links the KNOWN side and the WANTED side, write the equation and solve. In the tank problem, Wanjiru is 8 m away at 37 degrees, so $8 \times \tan 37 = 6.03$ m above eye level, and with her 1.5 m eye height the top of the tank is 7.5 m.	We learned that tan links opposite and adjacent, sin links opposite and hypotenuse, and cos links adjacent and hypotenuse. We learned that the ratio depends only on the angle, so one angle and one side are enough to find another side. We also learned to use $\tan 37 = 0.7536$ in DEG mode and to add the 1.5 m eye height.	We can now partly explain how the tank height can be found without climbing it: the known distance and the angle give a ratio that lets us calculate the height. We still need to explain how to measure the angle accurately, and to check that Otieno (10 m, 31 degrees) gets the same height, which we will test in later lessons.
Lesson 5 Angle of Elevation and Building the Clinometer	We built a clinometer from a cardboard protractor, a straw, a string and a washer. When the straw is level the string hangs at the 90° mark, and when we sighted the top of the classroom door frame (2.1 m high) from 3.0 m the string moved away from 90° by about 11°.	The angle of elevation is the angle above the horizontal from the observer's eye up to the object. On the clinometer it is the number of degrees between the string and the 90° mark, so the scale reading is 90° minus the angle of elevation (complementary angles). Height = (distance $\times \tan$ angle) + 1.5 m.	Wanjiru can get an angle for the Kilimani Secondary School water tank without climbing because her clinometer measures the angle of elevation from her eye. That angle, together with her 8 m horizontal distance, gives a right-angled triangle in which tan connects the two, and adding her 1.5 m eye height gives the tank's height.
Lesson 6 Angle of Depression and Outdoor Measurement: Measuring the Flagpole at Kilimani Secondary School	What did we observe? Record the veranda results (eye level 3.0 m, marker 4.0 m from the base, angle of depression about 37 degrees, with $4.0 \times \tan 37$ degrees = 3.01 m, about 3.0 m) and our flagpole data (distance, clinometer angle, calculated height, tape value 9.0 m).	What did we learn? Write what the angle of depression is, how it relates to the angle of elevation from the lower point (alternate angles), and how field data compare with the tape measurement (for Group Amani: 15 m, 27 degrees, 9.1 m versus 9.0 m).	How does this help explain the phenomenon? Explain, in your own words, how the height of the water tank can be found without climbing it, why the answers are close to the real height but not exact, and what causes the small error when we read angles to the nearest degree.
Lesson 7 Multi-Step Problems and Error Analysis: Which Ratio Do I Choose?	What did we observe? Record the three problems (Rift Valley cliff, Watamu diver, Nairobi ramp): the known side, the wanted side, the ratio we chose, and our answers (about 126 m, 9.2 m and 0.7 m). Add the four errors found in the critique task.	What did we learn? Write how to choose sin, cos or tan from the known and wanted sides, when to add the 1.5 m eye height, why DEG mode matters, and how our Lesson 1 predictions have changed.	How does this help explain the phenomenon? Explain in your own words why Wanjiru (8 m, 37 degrees) and Otieno (10 m, 31 degrees) both get a tank height of 7.5 m, and how choosing the correct ratio and avoiding common errors lets us find the height of the water tank without climbing it.
Lesson 8 Solving the Water Tank Problem: Final Explanation	What did we observe? Wanjiru (8 m from the tower, 37°) and Otieno (10 m, 31°) got different angles. Both calculated the same height: 7.53 m and 7.51 m, which round to 7.5 m. Mr. Kamau's tape measured 7.5 m. Group Amani's flagpole calculation gave 9.1 m against the tape's 9.0 m.	What did we learn? For a given angle, the ratio opposite $\div$ adjacent (tan) stays the same whatever the size of the triangle. A closer observer sees a larger angle. The height is found using height = (distance $\times \tan$ angle) + 1.5 m eye height. Angles read to the nearest degree cause small errors of about 0.1 m. A complete solution has a labelled diagram, the chosen ratio with a reason, the calculation, the answer with units and an explanation.	How can we explain the phenomenon? The tank height can be found without climbing because the tower, the ground and the line of sight form a right-angled triangle. We measure the distance along level ground and the angle with a clinometer, and tan of the angle links them to the height above eye level. We add 1.5 m eye height to get 7.5 m. Both learners agree because the ratio depends only on the angle. Trigonometry is important because it lets

			surveyors, engineers and foresters find heights and distances they cannot measure directly.
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